QUESTION BANK

SUBJECT :24MCASET104 – ADVANCED SOFTWARE ENGINEERING

SEM/YEAR: I/I

Module I

Introduction to Software Engineering: What is Software Engineering, Characteristics of Software, Life cycle of a software system: software design, development, testing, deployment, Maintenance, Project planning phase: project objectives, empirical estimation models, COCOMO, staffing and personnel planning. Software Engineering models: Predictive software engineering models, predictive and adaptive waterfall, waterfall with feedback (Sashimi), incremental waterfall, V model; Use cases and User stories.

Q No	Question	Competence
1.	Explain about the general phases in a Software Development Life cycle.	Understand
2.	What are the essential characteristics of software? Explain with suitable example	
3.	Discuss about the importance of cost estimation models in a software engineering process by taking the example of COCOMO.	Understand
4.	Assume that you are the project manager of a real-time critical project that involves hard deadlines. Which software process Model you will select and substantiate the answer.	Understand
5.	Define Software Engineering and its characteristics with suitable example.	Remember
6.	Explain the Life cycle of a software system in detail	Remember
7.	Explain predictive and adaptive waterfall model. How it is different from development phase models?	Remember
8.	Differentiate use cases and user stories with examples.	Remember
9.	Explain software requirement specifications with the help of an example	Remember
10.	Define predictive and adaptive development model in SDLC	Remember
11.	Explain about the various models used in SDLC	Remember
12.	Compare use case and user stories	Understand
13.	Explain about V-model and its features.	Remember

Module II Programming Style Guides and Coding Standards; Literate programming and Software documentation; Documentation generators, Javadoc Version control systems basic concepts; Concept of Distributed version control system and Git; Setting up Git; Core operations in Git version control system using command line interface (CLI): Clone a repository; View history; Modifying files; Branching; Push changes, Clone operation, add, commit, log. Pushing changes to the master; Using Git in IDEs and UI based tools. Software Quality: Understanding and ensuring requirements specification quality, design quality, quality in software development, conformance quality.		
Q No	**Question**	**Competence**
1.	Compare and contrast Git with Subversion (SVN) in terms of functionality and use cases.	Understand
2.	What is a version control system (VCS)? Compare centralized and distributed VCS.	Understand
3.	Discuss in details about software quality.	
4.	Discuss the concept of design quality in software engineering. How does it contribute to the overall quality of the software?	Remember
5.	Detail about Programming Style Guides and Coding Standards.	
6.	What is the purpose of version control system? List any four advantages of using Git.	Understand
7.	Explain the core operations in Git Version Control System to develop a good software product.	Remember
8.	Explain the four dimensions of software quality that decides the quality of a software product.	Understand
9.	How software quality is measured?	Understand
10.	Explain the differences between "git fetch" and "git pull". How can conflicts be resolved in git?	Understand
11.	Explain about version control system. How can we clone a git repository?	Understand
12.	Explain about the various comments used in git.	Understand
13.	Explain about the four dimensions of software quality.	Understand
14.	Explain about git and its features	Understand
15.	Explain about the four fundamental elements of git.	Understand
16.	Explain Literate Programming. How does it improve understanding and maintainability?	Understand

Module III		
OOP Concepts; Design Patterns: Basic concepts of Design patterns, How to select a design pattern, Creational patterns, Structural patterns, Behavioural patterns. Concept of Antipatterns. Unit testing and Unit Testing frameworks, The xUnit Architecture, Writing Unit Tests using at least one of Junit (for Java), unit test (for Python). Writing tests with Assertions, defining and using Custom Assertions, single condition tests, testing for expected errors, Abstract test.		

Q No	Question	Competence
1.	Explain the concept of encapsulation and how it improves software design.	Understand
2.	Explain the purpose of unit testing in software development. How does it contribute to overall software quality?	Understand
3.	What are design patterns? Describe the significance of reusable design solutions in software engineering.	
4.	Define anti-patterns and explain how they differ from design patterns? Why is it important for developers to recognize and avoid anti-patterns?	Understand
5.	Define the concept of Anti-patterns	Remember
6.	Define Unit testing. Explain the xUnit architecture with its components.	Understand
7.	Define Structural design pattern in detail.	Understand
8.	Explain about design patterns and its categories.	Understand
9.	Explain about behavioral design patterns in detail.	Understand
10.	Illustrate the importance of writing tests with assertions.	Application
11.	Explain the concept of Anti patterns.	Understand
12.	Define design patterns and its categories	Remember
13.	Explain about creational design patterns and its categories in detail.	Understand
14.	Explain Unit testing and Unit testing frameworks.	Understand
15.	Explain the role of asserting in unit testing and describe how custom assertions help improve test readability.Using JUnit,write a test case that verifies a divide() method correctly throws an exception when division by zero occurs.Use appropriate assertions.	

Module IV		
Concepts of Agile Development methodology; Scrum Framework. Software testing principles, Program walkthroughs, Program reviews; Blackbox testing: Equivalence class testing, Boundary value testing, White box testing: control flow testing, Data flow testing. Defect life cycle; Regression testing.		
Q No	**Question**	**Competence**
1.	Discuss about the regression testing in detail.	Understand
2.	"Software industry prefers the model like SCRUM nowadays." Justify the statement with adequate examples.	Understand
3.	Compare black box and white box testing.	Remember
4.	Explain the relevance of software testing.	Understand
5.	Explain the relevance of defect life cycle with a neat diagram	Understand
6.	List out the differences between program Inspection and program Walkthroughs.	Application
7.	Define product backlog in software engineering.	Remember
8.	Explain software testing principles in detail.	Understand
9.	Describe the various categories of error checklists for program inspections	Understand
10.	A program processes customer transactions and update account balance.using white box testing techniques,draw a control flow graph for any sampe logic.Identify the independent paths and design suitable test cases.Also explain how data flow anomalies may be detected for the same sample code	Understand
11.	What is black box testing and white box testing? Explain any two techniques used in black box testing and white box testing.	Remember
12.	Define the concept of software testing.	Remember
13.	Explain Equivalence class partitioning (ECP) and Boundary Value Partitioning (BVP).Apply both techniques to test an input field that accept age between 18 and 60.	Understand
14.	Explain Agile development and the Scrum framework. Describe the roles, ceremonies, and artifacts in Scrum, and explain how they help teams build software step by step.	Understand

Module V		
Software Configuration Management: Using version control, Managing dependencies, Continuous Integration: Prerequisites for continuous integration .Continuous Delivery: Principles of Software delivery. Build and deployment automation, Learn to use Ansible for configuration management. Test automation (as part of continuous integration), Learn to set up test automation cases using Robot Framework.		
Q No	**Question**	**Competence**
1.	Explain about the importance of automation in product delivery.	Understand
2.	Explain the architecture of Ansible. How do concepts like playbooks, inventory, and roles simplify configuration Management?	Understand
3.	Explain how the Robot Framework simplifies the creation of automated test cases. Provide an example of a basic test case using this framework.	Understand
4.	Describe the purpose of software configuration Management in software industry.	Understand
5.	What is CI/CD pipeline? Write the principles of software delivery.	Understand
6.	Describe the various strategies involved in continuous integration.	Remember
7.	Define the following process by using proper examples a) continuous integration b) continuous delivery	Remember
8.	Define test automation. Also compare build automation and deployment automation in software management	Remember
9.	Explain about principles of software delivery.	
10.	Explain the concept of Continuous integration and mention its prerequisites.	Remember
11.	Explain the concept of software	Understand